

Module 1: Probability Theory

(Based on Prof. Neeraj Misra notes and book Introduction to Mathematical Statistics by Hogg and Craig)

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1 Introduction

A motivating example:

Example 1.1. *The production manager of a bulb manufacturing company wishes to study the effect of a new manufacturing process on the lifetime of bulbs produced through it. Here the population under study is $\mathcal{P}$: collection of lifetimes of all electric bulbs produced using new manufacturing process.*

There are several other areas for instance, in medical research, interest may center on the effect of a drug that is to be administered; or an economist may be concerned with the prices of three specified commodities at various time intervals; or an agronomist may wish to study the effect that a chemical fertilizer has on the yield of a cereal grain.

In practice the population ($\mathcal{P}$) is generally large (e.g. collection of lifetimes of all elective bulbs that would be produced based on new manufacturing process) and it is not possible to (due to time/cost constraints) to get complete information about $\mathcal{P}$. Consequently, a representative sample (a sample that in certain sense is a true representative sample of the population) is taken from $\mathcal{P}$. Using this representative sample inferences regarding various population characteristics of $\mathcal{P}$ (such as population mean, population variance etc) are made.

X : lifetime of a typical electric bulb manufactured using new manufacturing process (a typical element of $\mathcal{P}$).

X is random (called a *random variable*) and its value varies across $\mathcal{P}$ according to some law.

Probability theory: A mathematical tool for modelling uncertainty (e.g. to describe the law according to which values of X vary across $\mathcal{P}$).

Statistics: Concerns with procedures for analysing data (sample) and drawing inferences about various characteristics of the population $\mathcal{P}$.

The only way to elicit information about any random phenomenon is to perform experiments (e.g. selecting a set of bulbs manufactured by the new manufacturing process and putting them on test for measuring the lifetimes). Each experiment terminates in an outcome which can not be predicted in advance prior to the performance of experiment.

Definition 1.1 (Random experiment). A random experiment is an experiment in which

- (a) all possible outcomes of the experiment are known in advance,
- (b) outcome of a particular trial of the experiment can not be predicted in advance,
- (c) the experiment can be repeated under identical conditions.

We shall use the notation $\mathcal{E}$ to denote a random experiment.

Definition 1.2. The collection of all possible outcomes of a random experiment is called its sample space.

We shall use the notation Ω to denote the sample space.

Example 1.2. $\mathcal{E}$: Tossing a coin once.
Sample space $\Omega = \{H, T\}$, where H : heads, T : tails.

Example 1.3. $\mathcal{E}$: Casting one red die and one white die.
Sample space $\Omega = \{(r, w) : r \text{ is the number of spots up on the red die, } w \text{ is the number of spots up on the white die}\} = \{(1, 1), (1, 2), \dots, (1, 6), (2, 1), \dots, (2, 6), \dots, (6, 1), \dots, (6, 6)\} = \{(i, j) : i, j \in \{1, 2, \dots, 6\}\} = \{1, 2, \dots, 6\} \times \{1, 2, \dots, 6\}$. It has 36 elements.

Example 1.4. $\mathcal{E}$: Putting two electric bulbs produced by new manufacturing process into test and measuring their lifetimes.
Sample space: $\Omega = \{(r_1, r_2) : r_1 \geq 0, r_2 \geq 0\} = [0, \infty]^2$.

Definition 1.3 (Events). If the outcome of a random experiment is a member of the set $E \subseteq \Omega$, we say that event E has occurred.

Generally we are interested in specific subsets of Ω , called events. Thus event space (events under consideration) $\mathcal{F}$ is a subset of power set Ω . In many situations $\mathcal{F} = P(\Omega)$ power set of Ω .

Example 1.5. In Example 1.3, $A = \{(1, 5), (6, 4)\}$ is an event.
In Example 1.4, $A = \{(r_1, r_2) : r_1 \leq 6, r_2 \geq 8\} = [0, 6] \times [8, \infty]$ can be an event.

2 Probability function (or probability measure)

Recall that $\mathcal{E}$: a random experiment, Ω : sample space of $\mathcal{E}$, and $\mathcal{F}$: event space. For all practical purposes one may take $\mathcal{F} = P(\Omega)$. A set function is a function whose domain is a collection of sets (called class of sets).

Definition 2.1 (Probability function or probability measure). A probability function is a real-valued set function, defined on an event space $\mathcal{F}$, satisfying the following axioms:

- (a) $\mathbb{P}(\Omega) = 1$;
- (b) $\mathbb{P}(E) \geq 0$, for all $E \in \mathcal{F}$,
- (c) If $E_1, E_2, \dots$, is a sequence of mutually exclusive/disjoint sets in $\mathcal{F}$ ($E_i \cap E_j = \emptyset$ for all $i \neq j$), then

$$\mathbb{P}(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^n \mathbb{P}(E_i) \quad (1)$$

Property (1) of the probability measure $\mathbb{P}$ is known as countable additivity. The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a probability space.

Remark 2.1. Axiom (b) and (c) are desirable for any measure (such as area; volume; probability, etc). Since the sample space Ω consists of all possible outcomes its occurrence is certain and therefore, Axiom (a) is also reasonable.

2.1 Elementary properties of probability measure

Let $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

Theorem 2.1. For each event $E \in \mathcal{F}$, $\mathbb{P}(E) = 1 - \mathbb{P}(E^c)$, where E^c is the complement of E defined as $E^c = \Omega \setminus E$.

Proof. We have $\Omega = E \cup E^c$ and $E \cap E^c = \emptyset$. Thus, by Axioms (a) and (c), it follows that

$$1 = \mathbb{P}(E) + \mathbb{P}(E^c),$$

this completes the proof. □

Corollary 2.1.1. *The probability of the null set is zero; that is, $\mathbb{P}(\emptyset) = 0$.*

Proof. In Theorem 2.1, take $E = \emptyset$ so that $E^c = \Omega$. Therefore,

$$\mathbb{P}(\emptyset) = 1 - \mathbb{P}(\Omega) = 1 - 1 = 0.$$

□

An alternate proof:

Lemma 1. $\mathbb{P}(\emptyset) = 0$.

Proof. Let $E_1 = \Omega$ and $E_i = \emptyset$ for $i = 2, 3, \dots$. Then $E_i \cap E_j = \emptyset$, for all $i \neq j$, and $\Omega = \cup_{i=1}^{\infty} E_i$. Therefore,

$$\begin{aligned} \mathbb{P}(\Omega) &= \mathbb{P}(\cup_{i=1}^{\infty} E_i) \\ 1 &= \sum_{i=1}^{\infty} \mathbb{P}(E_i) \quad (\text{Axiom (a) and (c)}) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(E_i) = \lim_{n \rightarrow \infty} [\mathbb{P}(\Omega) + (n-1)\mathbb{P}(\emptyset)] \\ &= 1 + \lim_{n \rightarrow \infty} [(n-1)\mathbb{P}(\emptyset)], \end{aligned}$$

Thus,

$$\mathbb{P}(\emptyset) = 0.$$

□

Theorem 2.2. *For any natural number n , let $E_1, \dots, E_n \in \mathcal{F}$ be mutually exclusive. Then*

$$\mathbb{P}(\cup_{i=1}^n E_i) = \sum_{i=1}^n \mathbb{P}(E_i).$$

Proof. Let $E_{n+1} = E_{n+2} = \dots = \emptyset$. Then $E_i \cap E_j = \emptyset$ for all $i \neq j$ and $\cup_{i=1}^n E_i = \cup_{i=1}^{\infty} E_i$. Thus,

$$\begin{aligned} \mathbb{P}(\cup_{i=1}^n E_i) &= \mathbb{P}(\cup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} \mathbb{P}(E_i) \quad (\text{by Axiom (c)}) \\ &= \sum_{i=1}^n \mathbb{P}(E_i), \quad (\mathbb{P}(E_i) = \mathbb{P}(\emptyset) = 0, \quad i = n+1, n+2, \dots) \end{aligned}$$

□

Theorem 2.3 (Inclusion–exclusion principle for two events). *Let $E_1, E_2 \in \mathcal{F}$. Then*

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2).$$

Proof. Observe that $E_1 \cup E_2 = E_1 \cup (E_2 - E_1)$ and $E_1 \cap (E_2 - E_1) = \emptyset$ (Venn diagrams will be effective to describe $E_2 - E_1$ and previous equalities). Next,

$$\begin{aligned} \mathbb{P}(E_1 \cup E_2) &= \mathbb{P}(E_1 \cup (E_2 - E_1)) \\ &= \mathbb{P}(E_1) + \mathbb{P}(E_2 - E_1), \quad (\text{Using Theorem 2.2}) \end{aligned} \tag{2}$$

Also, $E_2 = (E_1 \cap E_2) \cup (E_2 - E_1)$ and $(E_1 \cap E_2) \cap (E_2 - E_1) = \emptyset$. Thus,

$$\mathbb{P}(E_2) = \mathbb{P}((E_1 \cap E_2) \cup (E_2 - E_1)) = \mathbb{P}(E_1 \cap E_2) + \mathbb{P}(E_2 - E_1).$$

Therefore,

$$\mathbb{P}(E_2 - E_1) = \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2). \tag{3}$$

Using equations (2) and (3) we get

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2).$$

□

Theorem 2.4. For $E_1, \dots, E_k \in \mathcal{F}$, ($k \geq 2$ is an integer), let

$$p_{1,k} = \mathbb{P}(E_1) + \dots + \mathbb{P}(E_k) = \sum_{i=1}^k \mathbb{P}(E_i).$$

$$\begin{aligned} p_{2,k} &= \mathbb{P}(E_1 \cap E_2) + \dots + \mathbb{P}(E_1 \cap E_k) + \mathbb{P}(E_2 \cap E_3) + \dots + \mathbb{P}(E_2 \cap E_k) + \dots + \mathbb{P}(E_{k-1} \cap E_k) \\ &= \sum_{1 \leq i < j \leq k} \mathbb{P}(E_i \cap E_j) \end{aligned}$$

The above expression is the sum of probabilities of all possible intersections involving 2 events out of the k events $E_1, \dots, E_k$. Likewise

$$p_{i,k} = \sum_{1 \leq j_1 < j_2 < \dots < j_i \leq k} \mathbb{P}(E_{j_1} \cap E_{j_2} \cap \dots \cap E_{j_i})$$

The above expression is the sum of probabilities of all possible intersections involving i events out of k events $E_1, \dots, E_k$; $i = 1, \dots, k$. Then

$$\mathbb{P}(\cup_{i=1}^k E_i) = p_{1,k} - p_{2,k} + p_{3,k} + \dots + (-1)^{k-1} p_{k,k}. \quad (4)$$

Proof. Note that for $k = 2$, $p_{1,2} = \mathbb{P}(E_1) + \mathbb{P}(E_2)$, $p_{2,2} = \mathbb{P}(E_1 \cap E_2)$ and

$$\mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2) = p_{1,2} - p_{2,2}.$$

Thus the result is true for $k = 2$. Now suppose that the result is true for $k = 2, 3, \dots, m$, that is

$$\mathbb{P}(\cup_{i=1}^k E_i) = p_{1,k} - p_{2,k} + p_{3,k} + \dots + (-1)^{k-1} p_{k,k}; \quad \text{for all } k = 2, \dots, m.$$

Then

$$\begin{aligned} \mathbb{P}(\cup_{i=1}^{m+1} E_i) &= \mathbb{P}((\cup_{i=1}^m E_i) \cup E_{m+1}) = \mathbb{P}(\cup_{i=1}^m E_i) + \mathbb{P}(E_{m+1}) - \mathbb{P}((\cup_{i=1}^m E_i) \cap E_{m+1}) \\ &= \sum_{j=1}^m (-1)^{j-1} p_{j,m} + \mathbb{P}(E_{m+1}) - \sum_{j=1}^m (-1)^{j-1} t_{j,m}, \end{aligned}$$

where

$$\begin{aligned} t_{1,m} &= \sum_{i=1}^m \mathbb{P}(E_i \cap E_{m+1}) \\ t_{2,m} &= \sum_{1 \leq i < j \leq m} \mathbb{P}(E_i \cap E_j \cap E_{m+1}) \\ &\vdots \\ t_{j,m} &= \sum_{1 \leq i_1 < i_2 < \dots < i_j \leq m} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_j} \cap E_{m+1}); \quad j = 1, \dots, m \\ t_{m,m} &= \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_m \cap E_{m+1}). \end{aligned}$$

Therefore,

$$\begin{aligned} \mathbb{P}(\cup_{i=1}^{m+1} E_i) &= (p_{1,m} + \mathbb{P}(E_{m+1})) - (p_{2,m} + t_{1,m}) \\ &\quad + (p_{3,m} + t_{2,m}) + \dots + (-1)^{m-1} (p_{m,m} + t_{m-1,m}) + (-1)^m t_{m,m} \\ &= p_{1,m+1} - p_{2,m+1} + \dots + (-1)^{m-1} p_{m,m+1} + (-1)^m p_{m+1,m+1}, \end{aligned}$$

since

$$\begin{aligned}
p_{1,m} + \mathbb{P}(E_{m+1}) &= \sum_{j=1}^m \mathbb{P}(E_j) + \mathbb{P}(E_{m+1}) = p_{1,m+1} \\
p_{2,m} + t_{1,m} &= \sum_{1 \leq i < j \leq m} \mathbb{P}(E_i \cap E_j) + \sum_{i=1}^m \mathbb{P}(E_i \cap E_{m+1}) \\
&= \sum_{1 \leq i < j \leq m+1} \mathbb{P}(E_i \cap E_j) = p_{2,m+1}.
\end{aligned}$$

Likewise

$$\begin{aligned}
p_{m,m+1} + t_{m-1,m} &= \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_m) + \sum_{1 \leq i_1 < i_2 < \dots < i_{m-1} \leq m} \mathbb{P}(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_{m-1}} \cap E_{m+1}) \\
&= p_{m,m+1}
\end{aligned}$$

and

$$t_{m,m} = \mathbb{P}(E_1 \cap E_2 \cap \dots \cap E_m \cap E_{m+1}) = p_{m+1,m+1}.$$

The result now follows by induction. □

Exercise 2.1. Write the inclusion–exclusion principle for three events.

Theorem 2.5 (Boole’s inequality). *For some positive integer $k \geq 2$, let $E_1, \dots, E_k \in \mathcal{F}$. Then*

$$p_{1,k} - p_{2,k} \leq \mathbb{P}(\cup_{i=1}^k E_i) \leq p_{1,k}$$

Proof. Note that for $k = 2$,

$$\begin{aligned}
p_{1,2} &= \mathbb{P}(E_1) + \mathbb{P}(E_2) \\
p_{2,2} &= \mathbb{P}(E_1 \cap E_2) \\
\mathbb{P}(E_1 \cup E_2) &= \mathbb{P}(E_1) + \mathbb{P}(E_2) - \mathbb{P}(E_1 \cap E_2) \leq \mathbb{P}(E_1) + \mathbb{P}(E_2)
\end{aligned}$$

Thus

$$p_{1,2} - p_{2,2} = \mathbb{P}(E_1 \cup E_2) \leq \mathbb{P}(E_1) + \mathbb{P}(E_2).$$

Hence the result is true for $K = 2$. Now suppose that for some positive integer $m \geq 2$,

$$p_{1,k} - p_{2,k} \leq \mathbb{P}(\cup_{i=1}^k E_i) \leq p_{1,k}, \text{ for all } k = 1, \dots, m.$$

Then

$$\begin{aligned}
\mathbb{P}(\cup_{i=1}^{m+1} E_i) &= \mathbb{P}((\cup_{i=1}^m E_i) \cup E_{m+1}) \leq \mathbb{P}(\cup_{i=1}^m E_i) + \mathbb{P}(E_{m+1}) \\
&\leq p_{1,m} + \mathbb{P}(E_{m+1}) = p_{1,m+1}
\end{aligned} \tag{5}$$

Also using the result for $k = m$ we get

$$\mathbb{P}(\cup_{i=1}^m E_i) \geq p_{1,m} - p_{2,m}$$

and

$$\mathbb{P}(\cup_{i=1}^m (E_i \cap E_{m+1})) \leq \sum_{i=1}^m \mathbb{P}(E_i \cap E_{m+1}).$$

Thus

$$\begin{aligned}
\mathbb{P}(\cup_{i=1}^{m+1} E_i) &= \mathbb{P}((\cup_{i=1}^m E_i) \cup E_{m+1}) \\
&= \mathbb{P}(\cup_{i=1}^m E_i) + \mathbb{P}(E_{m+1}) - \mathbb{P}(\cup_{i=1}^m (E_i \cap E_{m+1})) \\
&\geq p_{1,m} - p_{2,m} + \mathbb{P}(E_{m+1}) - \sum_{i=1}^m \mathbb{P}(E_i \cap E_{m+1}) \\
&= (p_{1,m} + \mathbb{P}(E_{m+1})) - (p_{2,m} + \sum_{i=1}^m \mathbb{P}(E_i \cap E_{m+1})) \\
&= p_{1,m+1} - p_{2,m+1}.
\end{aligned} \tag{6}$$

Using (5) and (6) we get

$$p_{1,m+1} - p_{2,m+1} \leq \mathbb{P}(\cup_{i=1}^{m+1} E_i) \leq p_{1,m+1}$$

and the result follows using principle of mathematical induction. $\square$

Remark 2.2. It can be shown that

$$\begin{aligned} p_{1,k} - p_{2,k} + p_{3,k} - p_{4,k} &\leq \mathbb{P}(\cup_{i=1}^k E_i) \leq p_{1,k} - p_{2,k} + p_{3,k} \\ &\vdots \\ p_{1,k} - p_{2,k} + p_{3,k} - \dots - p_{2m+k} &\leq \mathbb{P}(\cup_{i=1}^k E_i) \leq p_{1,k} - p_{2,k} + p_{3,k} - \dots + p_{2m-1,k} \end{aligned}$$

where $m = 1, 2, \dots, \lfloor \frac{k}{2} \rfloor$.

Theorem 2.6 (Bonferroni's inequality). *Let $E_1, \dots, E_k \in \mathcal{F}$. Then*

$$\mathbb{P}(\cap_{i=1}^k E_i) \geq \max\{\sum_{i=1}^k \mathbb{P}(E_i) - (k-1), 0\}.$$

Proof. We have

$$\begin{aligned} \mathbb{P}(\cap_{i=1}^k E_i) &= \mathbb{P}((\cup_{i=1}^k E_i^c)^c), \quad \text{by De-Morgan's law} \\ &= 1 - \mathbb{P}(\cup_{i=1}^k E_i^c) \geq 1 - \sum_{i=1}^k \mathbb{P}(E_i^c) \quad \text{by Boole's inequality} \\ &= 1 - \sum_{i=1}^k [1 - \mathbb{P}(E_i)] = \sum_{i=1}^k \mathbb{P}(E_i) - (k-1). \end{aligned} \tag{7}$$

Also

$$\mathbb{P}(\cap_{i=1}^k E_i) \geq 0 \tag{8}$$

Combining (7) and (8) we get

$$\mathbb{P}(\cap_{i=1}^k E_i) \geq \max\{\sum_{i=1}^k \mathbb{P}(E_i) - (k-1), 0\}.$$

$\square$

Example 2.1. Random experiment: Casting a red and white die

Sample space: $\Omega = \{(i, j) : i, j \in \{1, 2, \dots, 6\}\}$. For $(i, j) \in \Omega$, i : number of spots up on the red die; j : number of spots up on the white die. *Event space* $\mathcal{F}$ = power set of Ω . For $E \in \mathcal{F}$, define $Q : \mathcal{F} \rightarrow \mathbb{R}$ as

$$Q(E) = \frac{|E|}{36}, \quad \text{where } |E| = \text{number of elements in } E.$$

Then

$$(a) \quad Q(\Omega) = \frac{|\Omega|}{36} = \frac{36}{36} = 1.$$

$$(b) \quad Q(E) = \frac{|E|}{36} \geq 0, \quad \text{for all } E \in \Omega.$$

(c) For mutually exclusive events $E_1, E_2, \dots$

$$\begin{aligned} Q(\cup_{i=1}^{\infty} E_i) &= \frac{|\cup_{i=1}^{\infty} E_i|}{36} = \frac{\sum_{i=1}^{\infty} |E_i|}{36}, \quad E_i \text{'s are disjoint} \\ &= \sum_{i=1}^{\infty} \frac{|E_i|}{36} = \sum_{i=1}^{\infty} Q(E_i). \end{aligned}$$

Thus $(\Omega, Q, \mathcal{F})$ is a probability space.

2.2 Equally likely probability models for finite sample space

Suppose that the sample space $\Omega = \{w_1, \dots, w_k\}$ is finite (has k elements). Here singletons $\{w_i\}$'s are called elementary events and $\Omega = \cup_{i=1}^k \{E_i\}$. Suppose that

$$\mathbb{P}(\{w_i\}) = \frac{1}{k},$$

that is each elementary event is equally likely.

For any event $E \subseteq \Omega$, we have $E = \{w_{i_1}, \dots, w_{i_r}\}$ for $i_1, \dots, i_r \in \{1, 2, \dots, k\}$, $1 \leq r \leq k$. Then $E = \cup_{j=1}^r \{w_{i_j}\}$,

$$\begin{aligned} \mathbb{P}(E) &= \mathbb{P}(\cup_{j=1}^r \{w_{i_j}\}) = \sum_{j=1}^r \mathbb{P}(\{w_{i_j}\}) = \sum_{i=1}^r \frac{1}{k} = \frac{r}{k} \\ &= \frac{\text{number of ways favourable for event } E}{\text{total number of ways in which the random experiment can terminate}} \end{aligned}$$

Here the assumption of equally likely ($\mathbb{P}(\{w_i\}) = 1/k$, $i = 1, \dots, k$) is a part of probability modeling.

“At random”: In a random experiment with finite sample space Ω whenever we say that the experiment has been performed at random it means that all the outcomes in the sample space are equally likely.

Example 2.2. *Birthday problem: Suppose that a college has n students including you. Each of them were born on non-leap years.*

- Find the probability that at least two of them have the same birthday. For what values of n this probability is more than 0.5, 0.8, 0.95.
- For what value of n the probability that you will find someone who share your birthday is $1/2$.

Solution:

- Probability that at least two students share a birthday.**

It is easier to compute the complement. The probability that all n students have distinct birthdays is

$$P(\text{all distinct}) = \frac{365}{365} \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365 - n + 1}{365} = \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

Hence, the probability that at least two students share the same birthday is

$$P(\text{at least one common birthday}) = 1 - \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

Numerical evaluation gives:

Probability	Smallest n
> 0.5	23
> 0.8	34
> 0.95	47

- Probability that someone shares your birthday.**

Fix your birthday. The probability that a given other student does not share your birthday is $364/365$. Therefore, the probability that none of the remaining $n - 1$ students shares your birthday is

$$P(\text{no one shares your birthday}) = \left(\frac{364}{365}\right)^{n-1}.$$

Thus,

$$P(\text{someone shares your birthday}) = 1 - \left(\frac{364}{365}\right)^{n-1}.$$

We want

$$1 - \left(\frac{364}{365}\right)^{n-1} = \frac{1}{2}.$$

Solving,

$$n - 1 = \frac{\ln(1/2)}{\ln(364/365)} \approx 253,$$

so

$$\boxed{n \approx 254}.$$

Example 2.3. Five cards are drawn at random and without replacement from a deck of 52 cards. Find the probability that:

(i) each card is spade (event E_1)

(ii) at least one card is spade (event E_2)

(iii) exactly two cards are king and two cards are queen (event E_3)

(iv) exactly two kings, two queens and one jack are drawn (event E_4).

Solution:

(i)

$$\mathbb{P}(E_1) = \frac{\binom{13}{5}}{\binom{52}{5}}.$$

(ii)

$$\begin{aligned}\mathbb{P}(E_2) &= 1 - \mathbb{P}(E_2^c) = 1 - \mathbb{P}(\text{no card is spade}) \\ &= 1 - \frac{\binom{39}{5}}{\binom{52}{5}}\end{aligned}$$

(iii)

$$\mathbb{P}(E_3) = \frac{\binom{4}{3} \binom{4}{2}}{\binom{52}{5}}$$

(iv)

$$\mathbb{P}(E_4) = \frac{\binom{4}{2} \binom{4}{2} \binom{4}{1}}{\binom{52}{5}}$$

Example 2.4 (Capture/recapture method for estimating population size). In a wildlife population suppose that the population size n is unknown. To estimate the population size n , 20 animals are captured, tagged and then released back. Thereafter 40 animals are captured at random again and it is found that 8 of them are tagged. Find the estimate of population size n based on the given data.

Solution: We have n = total number of animals; 20 = number of tagged animals in the population; $n - 20$ = number of untagged animals in the population.

Data: Sample of 40 animals yield: number of tagged animals = 8 and number of untagged animals = 32.

The probability of obtaining this data is

$$\begin{aligned}
l(n) &= \frac{\binom{20}{8} \binom{n-20}{32}}{\binom{n}{40}}, \quad n \geq 52 \\
l(n+1) > l(n) &\iff \frac{\binom{n-19}{32}}{\binom{n+1}{40}} > \frac{\binom{n-20}{32}}{\binom{n}{40}} \\
&\iff \frac{n-19}{(n-51)(n+1)} > 1 \iff n < 99
\end{aligned}$$

Similarly,

$$l(n+1) < l(n) \iff n > 99.$$

Thus $l()$ is maximized at $n = 99$, i.e., for $n = 99$ the observed data (among 40 captured animals 8 are tagged and 32 are untagged) is most probable. Thus the estimate of n is 99 (maximum likelihood estimator).

2.3 Conditional probability

Consider a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, where $\Omega = \{w_1, \dots, w_n\}$ is finite and

$$\mathbb{P}(\{w_i\}) = \frac{1}{n}, \quad i = 1, \dots, n, \quad (\text{equally likely probability model}).$$

Then, for any event $C \in \mathcal{F}$,

$$\mathbb{P}(C) = \frac{\text{number of cases favourable to } C}{\text{total number of cases}} = \frac{|C|}{|\Omega|} = \frac{|C|}{n}.$$

Now suppose it is known apriori that event A has occurred (that is outcome of the experiment is an element of A), where $|A| \geq 1$ (so that $\mathbb{P}(A) = (|A|/n) > 0$). Given this prior information (that the event A has occurred), we want to define probability function say $\mathbb{P}(B|A)$ on event space $\mathcal{F}$. A natural way to define $\mathbb{P}(B|A)$ is

$$\mathbb{P}(B|A) = \frac{|A \cap B|}{|A|} = \frac{|A \cap B|/n}{|A|/n} = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}, \quad B \in \mathcal{F}.$$

Definition 2.2. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A \in \mathcal{F}$ be such that $\mathbb{P}(A) > 0$. Then

$$\mathbb{P}(B|A) = \frac{|A \cap B|}{|A|}, \quad B \in \mathcal{F}$$

is called the conditional probability of event B given the event A .

Remark 2.3. (a) In the above definition the event A (with $\mathbb{P}(A) > 0$) is fixed and for this fixed $A \in \mathcal{F}$, $\mathbb{P}(\cdot|A)$ is a set function defined on $\mathcal{F}$.

(b) $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A) = \mathbb{P}(B)\mathbb{P}(A|B)$, $A, B \in \mathcal{F}$.

Theorem 2.7. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$, be fixed. Then $\mathbb{P}(\cdot|A) : \mathcal{F} \rightarrow \mathbb{R}$ is a probability function (called the conditional probability function) on $\mathcal{F}$ (so that $(\Omega, \mathcal{F}, \mathbb{P}(\cdot|A))$ is a probability space).

Proof. Clearly,

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} \geq 0, \quad \text{for all } B \in \mathcal{F}.$$

and

$$\mathbb{P}(\Omega|A) = \frac{\mathbb{P}(A \cap \Omega)}{\mathbb{P}(A)} = 1.$$

Let $\{B_n\}_{n \geq 1}$ be a sequence of disjoint events in $\mathcal{F}$. Then

$$\mathbb{P}(\cup_{n=1}^{\infty} B_n | A) = \frac{\mathbb{P}((\cup_{n=1}^{\infty} B_n) \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(\cup_{n=1}^{\infty} (B_n \cap A))}{\mathbb{P}(A)}$$

Since $\{B_n\}_{n \geq 1}$ are disjoint then subsets $\{B_n \cap A\}_{n \geq 1}$ are also disjoint. Since $P(\cdot)$ is a probability measure, we get $\square$

$$\mathbb{P}(\cup_{n=1}^{\infty} B_n | A) = \frac{\sum_{n=1}^{\infty} \mathbb{P}((B_n \cap A))}{\mathbb{P}(A)} = \sum_{n=1}^{\infty} \frac{\mathbb{P}((B_n \cap A))}{\mathbb{P}(A)} = \sum_{n=1}^{\infty} \mathbb{P}(B_n | A).$$

It follows that $\mathbb{P}(\cdot|A)$ is a probability function on $\mathcal{F}$, for any fixed $A \in \mathcal{F}$ with $\mathbb{P}(A) > 0$.

Example 2.5. Five cards are drawn at random (without replacement) from a deck of 52 cards. Define events B : all spade in hand; A : at least 4 spades in hand. Find $\mathbb{P}(B|A)$.

Solution: We have

$$\begin{aligned} \mathbb{P}(B|A) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(B)}{\mathbb{P}(A)} \\ &= \frac{\binom{13}{5} / \binom{52}{5}}{[(\binom{13}{4} \binom{39}{1}) + \binom{13}{5}] / \binom{52}{5}}. \end{aligned}$$

Remark 2.4 (Multiplication Law). (i) $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)$, if $\mathbb{P}(A) > 0$.

(ii) $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A \cap B)\mathbb{P}(C|A \cap B) = \mathbb{P}(A)\mathbb{P}(B|A)\mathbb{P}(C|A \cap B)$, provided $\mathbb{P}(A \cap B) > 0$ (which ensures that $\mathbb{P}(A) > 0$).

(iii) Using the principle of mathematical induction, we have

$$\begin{aligned} \mathbb{P}(\cap_{i=1}^n C_i) &= \mathbb{P}(C_1 \cap C_2 \cap \dots \cap C_n) = \mathbb{P}(C_1 \cap C_2 \cap \dots \cap C_{n-1})\mathbb{P}(C_n | C_1 \cap C_2 \cap \dots \cap C_{n-1}) \\ &= \mathbb{P}(C_1 \cap C_2 \cap \dots \cap C_{n-2})\mathbb{P}(C_{n-1} | C_1 \cap C_2 \cap \dots \cap C_{n-2})\mathbb{P}(C_n | C_1 \cap C_2 \cap \dots \cap C_{n-1}) \\ &= \mathbb{P}(C_1)\mathbb{P}(C_2|C_1)\mathbb{P}(C_3|C_1 \cap C_2) \dots \mathbb{P}(C_n|C_1 \cap C_2 \cap \dots \cap C_{n-1}) \end{aligned}$$

provided $\mathbb{P}(C_1 \cap C_2 \cap \dots \cap C_{n-1}) > 0$ (which also ensures that $\mathbb{P}(C_1 \cap C_2 \cap \dots \cap C_i) > 0$, $i = 1, \dots, n-2$). Due to symmetry, if $(\alpha_1, \dots, \alpha_n)$ is any permutation of $(1, \dots, n)$ then

$$\begin{aligned} \mathbb{P}(\cap_{i=1}^n C_i) &= \mathbb{P}(C_{\alpha_1} \cap C_{\alpha_2} \dots \cap C_{\alpha_n}) \\ &= \mathbb{P}(C_{\alpha_1})\mathbb{P}(C_{\alpha_2}|C_{\alpha_1})\mathbb{P}(C_{\alpha_3}|C_{\alpha_1} \cap C_{\alpha_2}) \dots \mathbb{P}(C_{\alpha_n}|C_{\alpha_1} \cap \dots \cap C_{\alpha_{n-1}}) \end{aligned}$$

provided $\mathbb{P}(C_{\alpha_1} \cap \dots \cap C_{\alpha_{n-1}}) > 0$ (which also ensures that $\mathbb{P}(C_{\alpha_1} \cap \dots \cap C_{\alpha_i}) > 0$, $i = 1 \dots, n-1$).

Example 2.6. A bowl contains 3 red and 5 blue balls. All balls that are of the same colour are identical. Two balls are drawn successively at random and without replacement. Define events

A : first draw resulted in a red ball

B : second draw resulted in a blue ball.

Find $\mathbb{P}(A \cap B)$, $\mathbb{P}(A)$ and $\mathbb{P}(B)$.

Solution: There are 3 red and 5 blue balls, hence 8 balls in total.

Probability of A. The probability that the first draw is red is

$$\mathbb{P}(A) = \frac{3}{8}.$$

Probability of $A \cap B$. If the first draw is red, then 2 red and 5 blue balls remain, i.e. 7 balls in total. Hence,

$$\mathbb{P}(B \mid A) = \frac{5}{7}.$$

Therefore,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}.$$

Probability of B. Using the law of total probability,

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

Now $\mathbb{P}(A^c) = \frac{5}{8}$ and, if the first draw is blue, then 4 blue balls remain out of 7, so

$$\mathbb{P}(B \mid A^c) = \frac{4}{7}.$$

Hence,

$$\mathbb{P}(B) = \frac{5}{7} \cdot \frac{3}{8} + \frac{4}{7} \cdot \frac{5}{8} = \frac{35}{56} = \frac{5}{8}.$$

Final answers:

$$\boxed{\mathbb{P}(A) = \frac{3}{8}, \quad \mathbb{P}(A \cap B) = \frac{15}{56}, \quad \mathbb{P}(B) = \frac{5}{8}.}$$

Definition 2.3 (Countable Set). A set Λ is said to be a countable set if its elements can either be put in one-one correspondence with $\mathbb{N} = \{1, 2, \dots\}$ or with $\{1, 2, \dots, n\}$, for some $n \in \mathbb{N}$.

Theorem 2.8. For a countable set Λ let $\{E_\alpha : \alpha \in \Lambda\}$ be a countable collection of mutually exclusive (that is $E_\alpha \cap E_\beta = \emptyset$, for all $\alpha \neq \beta$) and exhaustive (that is $\mathbb{P}(\cup_{\alpha \in \Lambda} E_\alpha) = 1$) events. Then for any $E \in \mathcal{F}$,

$$\mathbb{P}(E) = \sum_{\alpha \in \Lambda} \mathbb{P}(E \cap E_\alpha) = \sum_{\alpha \in \Lambda, \mathbb{P}(E_\alpha) > 0} \mathbb{P}(E|E_\alpha)\mathbb{P}(E_\alpha).$$

Proof. Since $\mathbb{P}(\cup_{\alpha \in \Lambda} E_\alpha) = 1$, we have

$$\begin{aligned} \mathbb{P}(E) &= \mathbb{P}(E \cap (\cup_{\alpha \in \Lambda} E_\alpha)) = \mathbb{P}(\cup_{\alpha \in \Lambda} (E \cap E_\alpha)) \\ &= \sum_{\alpha \in \Lambda} \mathbb{P}(E \cap E_\alpha) \quad \text{since } E_\alpha \text{ are disjoint} \\ &= \sum_{\alpha \in \Lambda, \mathbb{P}(E_\alpha) > 0} \mathbb{P}(E \cap E_\alpha) \quad (\mathbb{P}(E_\alpha) = 0 \implies \mathbb{P}(E \cap E_\alpha) = 0) \\ &= \sum_{\alpha \in \Lambda, \mathbb{P}(E_\alpha) > 0} \mathbb{P}(E|E_\alpha)\mathbb{P}(E_\alpha). \end{aligned}$$

□

Example 2.7. A population comprises of 40% female and 60% male. Suppose that 15% of female and 30% of male in the population smoke. A person is selected at random from the population.

- (a) Find the probability that he/she is a smoker.
- (b) Given that selected person is smoker, find the probability that he is male.

Solution: Define the events

$$\begin{aligned} M &: \text{selected person is a male} \\ F = M^c &: \text{selected person is a female} \\ S &: \text{selected person is a smoker} \\ T = S^c &: \text{selected person is a non-smoker} \end{aligned}$$

We have $\mathbb{P}(M) = 0.6$, $\mathbb{P}(F) = 0.4$, $\mathbb{P}(S|F) = 0.15$, $\mathbb{P}(T|F) = 0.85$, $\mathbb{P}(S|M) = 0.3$, and $\mathbb{P}(T|M) = 0.7$.

(a) By Theorem of total probability

$$\begin{aligned} \mathbb{P}(S) &= \mathbb{P}(S \cap F) + \mathbb{P}(S \cap M) = \mathbb{P}(S|F)\mathbb{P}(F) + \mathbb{P}(S|M)\mathbb{P}(M) \\ &= 0.15 \times 0.4 + 0.3 \times 0.6 = 0.06 + 0.18 = 0.24. \end{aligned}$$

(b)

$$\begin{aligned} \mathbb{P}(M|S) &= \frac{\mathbb{P}(M \cap S)}{\mathbb{P}(S)} = \frac{\mathbb{P}(S|M)\mathbb{P}(M)}{\mathbb{P}(S)} \\ &= \frac{0.3 \times 0.6}{0.24} = \frac{0.18}{0.24} = \frac{3}{4}. \end{aligned}$$

Theorem 2.9 (Bayes' theorem). Let $\{E_\alpha : \alpha \in \Lambda\}$ be a countable collection of mutually exclusive and exhaustive events and let E be any event with $\mathbb{P}(E) > 0$. Then for any $j \in \Lambda$ with $\mathbb{P}(E_j) > 0$,

$$\mathbb{P}(E_j|E) = \frac{\mathbb{P}(E_j)\mathbb{P}(E|E_j)}{\sum_{\alpha \in \Lambda, \mathbb{P}(E_\alpha) > 0} \mathbb{P}(E_\alpha)\mathbb{P}(E|E_\alpha)}. \quad (9)$$

Proof. For $j \in \Lambda$,

$$\mathbb{P}(E_j|E) = \frac{\mathbb{P}(E_j \cap E)}{\mathbb{P}(E)} = \frac{\mathbb{P}(E_j)\mathbb{P}(E|E_j)}{\sum_{\alpha \in \Lambda, \mathbb{P}(E_\alpha) > 0} \mathbb{P}(E_\alpha)\mathbb{P}(E|E_\alpha)}, \quad (\text{using Theorem of total probability}).$$

□

Remark 2.5. In Bayes theorem $\{\mathbb{P}(E_j) : j \in \Lambda\}$ are called prior probabilities and $\{\mathbb{P}(E_j|E) : j \in \Lambda\}$ are called posterior probabilities.

Example 2.8.

- Bowl C_1 contains 3 red and 7 blue chips;
- Bowl C_2 contains 8 red and 2 blue chips;
- Bowl C_3 contains 5 red and 5 blue chips.

All chips of the same colour are identical. A die is cast and a bowl is selected as per the following scheme:

- Bowl C_1 is selected if 5 or 6 spots show on the upper side;
- Bowl C_2 is selected if 2, 3, or 4 spots show on the upper side;
- Bowl C_3 is selected if 6 spots show on the upper face.

Selected bowl is handed over to another person who draws two chips at random from this bowl. Find the probability that:

- two red chips are drawn;
- given that drawn chips are both red, find the probability that it come from bowl C_3 .

Solution: Define the events

- A_i : Selected bowl is C_i , $i = 1, 2, 3$;
- R : the chips drawn from the selected bowl are both red.

Then

$$\mathbb{P}(A_1) = \frac{2}{6} = \frac{1}{3}; \quad \mathbb{P}(A_2) = \frac{3}{6} = \frac{1}{2}; \quad \mathbb{P}(A_3) = \frac{1}{6}.$$

$\{A_1, A_2, A_3\}$ are mutually exclusive and exhaustive.

(a)

$$\begin{aligned} \mathbb{P}(R) &= \mathbb{P}(R|A_1)\mathbb{P}(A_1) + \mathbb{P}(R|A_2)\mathbb{P}(A_2) + \mathbb{P}(R|A_3)\mathbb{P}(A_3) \\ &= \frac{\binom{3}{2}}{\binom{10}{2}} \times \frac{1}{3} + \frac{\binom{8}{2}}{\binom{10}{2}} \times \frac{1}{2} + \frac{\binom{5}{2}}{\binom{10}{2}} \times \frac{1}{6} \\ &= \frac{1}{6} \left[\frac{2 \times 3}{45} + \frac{3 \times 28}{45} + \frac{10}{45} \right] \\ &= \frac{1}{6} \times \frac{100}{45} = \frac{10}{27}. \end{aligned}$$

(b)

$$\mathbb{P}(A_3|R) = \frac{\mathbb{P}(R|A_3)\mathbb{P}(A_3)}{\mathbb{P}(R)} = \frac{\left[\binom{5}{2} / \binom{10}{2} \right] \times \frac{1}{6}}{10/27} = \frac{1}{10}.$$

Definition 2.4. Let $\{E_j : j \in \Lambda\}$ be a collection of events.

- (a) Events $\{E_j : j \in \Lambda\}$ are said to be pairwise independent if for any pair of events, E_i and E_j ($i, j \in \Lambda$, $i \neq j$) in the collection $\{E_j : j \in \Lambda\}$ we have

$$\mathbb{P}(E_i \cap E_j) = \mathbb{P}(E_i)\mathbb{P}(E_j).$$

- (b) Events $\{E_1, \dots, E_n\}$ are said to be independent if for any sub-collection $\{E_{\alpha_1}, \dots, E_{\alpha_k}\}$ of $\{E_1, \dots, E_n\}$ ($k = 2, 3, \dots, n$)

$$\mathbb{P}(\cap_{j=1}^k E_{\alpha_j}) = \prod_{j=1}^k \mathbb{P}(E_{\alpha_j}).$$

- (c) Events $\{E_\alpha : \alpha \in \Lambda \subseteq \mathbb{R}\}$ are said to be independent if any finite sub-collection of events in $\{E_\alpha : \alpha \in \Lambda \subseteq \mathbb{R}\}$ forms a collection of independent events.

Theorem 2.10. Let $E_1, E_2, \dots$ be a collection of independent events. Then

$$\mathbb{P}(\cap_{k=1}^{\infty} A_k) = \prod_{k=1}^{\infty} \mathbb{P}(A_k).$$

Proof. Let

$$B_n = \cap_{k=1}^n A_k, \quad n = 1, 2, \dots$$

Then $B_n \downarrow$. Using Problem 5(b), Assignment 1,

$$\mathbb{P}(\cap_{n=1}^{\infty} B_n) = \lim_{n \rightarrow \infty} \mathbb{P}(B_n).$$

But $\cap_{n=1}^{\infty} B_n = \cap_{k=1}^{\infty} A_k$ and

$$\mathbb{P}(B_n) = \mathbb{P}(\cap_{k=1}^n A_k) = \prod_{k=1}^n \mathbb{P}(A_k).$$

Then

$$\mathbb{P}(\cap_{k=1}^{\infty} A_k) = \lim_{n \rightarrow \infty} \prod_{k=1}^n \mathbb{P}(A_k) = \prod_{k=1}^{\infty} \mathbb{P}(A_k).$$

□

Remark 2.6.

- (a) To verify that n events $E_1, \dots, E_n$ are independent, one must verify

$$\binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} = 2^n - n - 1$$

conditions. For example to calculate that three events E_1, E_2 and E_3 are independent the following 4 ($2^3 - 3 - 1$) conditions must be verified:

- (i) $\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2)$;
- (ii) $\mathbb{P}(E_1 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_3)$;
- (iii) $\mathbb{P}(E_2 \cap E_3) = \mathbb{P}(E_2)\mathbb{P}(E_3)$;
- (iv) $\mathbb{P}(E_1 \cap E_2 \cap E_3) = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3)$.

- (b) Any sub-collection of independent events is of independent events. In practice the independence of a collection of events implies their pairwise independence.

- (c) If E_1 and E_2 are independent events ($\mathbb{P}(E_1) > 0, \mathbb{P}(E_2) > 0$), then

$$\mathbb{P}(E_1|E_2) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)} = \frac{\mathbb{P}(E_1)\mathbb{P}(E_2)}{\mathbb{P}(E_2)} = \mathbb{P}(E_1)$$

that is conditional probability of E_1 given E_2 is same as unconditional probability of E_1 .

Example 2.9. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with $\Omega = \{1, 2, 3, 4\}$ and $\mathbb{P}(\{i\}) = \frac{1}{4}$, $i = 1, 2, 3, 4$. Let $A = \{1, 4\}$, $B = \{2, 4\}$ and $C = \{3, 4\}$.

Solution: We have

$$\begin{aligned} \mathbb{P}(A) &= \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2} \\ \mathbb{P}(A \cap B) &= \mathbb{P}(B \cap C) = \mathbb{P}(A \cap C) = \frac{1}{4}. \end{aligned}$$

Thus

- (i) $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \frac{1}{4}$,
- (ii) $\mathbb{P}(A \cap C) = \mathbb{P}(A)\mathbb{P}(C) = \frac{1}{4}$,
- (iii) $\mathbb{P}(B \cap C) = \mathbb{P}(B)\mathbb{P}(C) = \frac{1}{4}$.

It implies that A, B and C are pairwise independent. However,

$$\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{4\}) = \frac{1}{4} \neq \frac{1}{8} = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C).$$

It implies that A, B and C are not independent although they are pairwise independent.

Example 2.10. Let $E_1, \dots, E_n$ be a collection of independent events. Show that,

- (a) For any permutation $(\alpha_1, \dots, \alpha_n)$ of $(1, \dots, n)$, $E_{\alpha_1}, \dots, E_{\alpha_n}$ are independent;
- (b) $E_1, \dots, E_k, E_{k+1}^c, \dots, E_n^c$ are independent for any $k \in \{1, 2, \dots, n-1\}$;
- (c) E_1^c and $E_2 \cup E_3^c \cup E_5$ are independent;
- (d) $E_1 \cup E_2^c, E_3^c$ and $E_4 \cap E_5^c$ are independent.